Do Now · retrieval

Lesson 130 · Symmetric Encryption: One Shared Secret

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the main role of a dictionary token in compression?

Q2

What is O(n) search complexity associated with?

Q3

In a BST, which side holds a smaller value?

Q4

In a queue, what does FIFO stand for?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start plaintext — Sender has message and shared secret key.

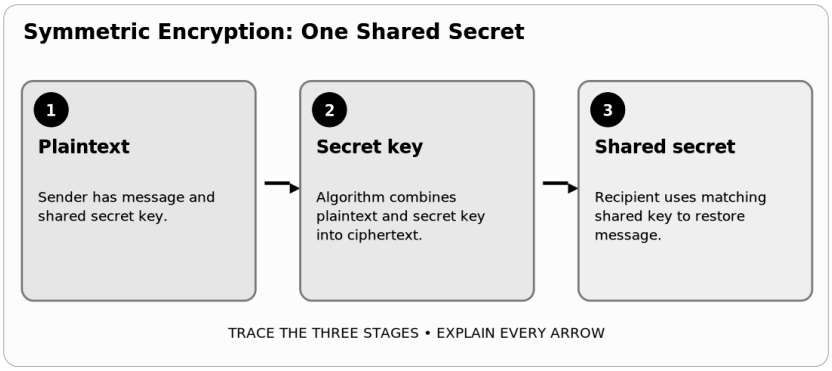
Core idea 2

Encrypt — Algorithm combines plaintext and secret key into ciphertext.

Core idea 3

Decrypt — Recipient uses matching shared key to restore message.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

plaintext — Readable original message.

ciphertext — Encrypted output.

symmetric key — Same secret key is used for encryption and decryption.

key distribution — Securely sharing the secret with authorised parties.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Alice and Bob share secret key K. Explain why intercepting the ciphertext is not enough to read the message.

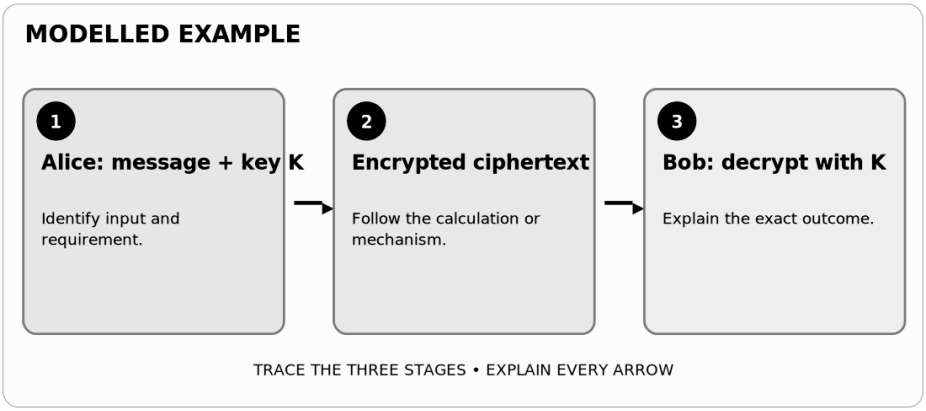
Follow the worked method

1. Alice: message + key K — Predict the first step from the prompt.

2. Encrypted ciphertext — Explain the intermediate result or decision.

3. Bob: decrypt with K — A secure symmetric cipher transforms plaintext using key K. Bob decrypts with the same key; an interceptor without K should not recover plaintext feasibly. Security requires a strong cipher and protected key.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Sensitive backup
Predict the first step from the prompt.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe the roles of plaintext, key and ciphertext in symmetric encryption. [6]

Question 2

Explain the main key-management challenge in sharing encrypted records between two offices. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate symmetric encryption for a large school backup stored with a third-party provider. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Symmetric encryption uses one public key for encryption and a separate private key for decryption.

Use key vocabulary from memory

1. Explain plaintext in your own words.

2. Explain ciphertext in your own words.

3. Explain symmetric key in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.